The Role of Nuclear Them1 in Metabolic Gene Regulation

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Abstract

Non-shivering thermogenesis is the process by which mammals produce heat. It is regulated by the protein thioesterase superfamily member 1 (Them1), which forms biomolecular condensates (puncta) to limit energy expenditure. In response to β_3-adrenergic stimulation, Them1 is phosphorylated, prompting its diffusion throughout the cytoplasm. Binding of lysophosphatidylcholine to diffuse Them1 induces a conformational change that allows for nuclear translocation. Previous studies have shown that nuclear Them1 interacts with transcription factors regulating *de novo* lipogenesis and glycolysis. However, there is still limited insight into its broader impact on overall gene expression. To address this, we examined the expression of thermogenesis-related genes in brown adipocytes using bulk RNA-seq validated by RT-qPCR to investigate the function of nuclear Them1. Here we show that nuclear Them1 influences gene expression across pathways related to fatty acid oxidation, mitochondrial function, glycolysis, and cholesterol biosynthesis. In addition, we found consistent downregulation of genes involved in thermogenesis, lipogenesis, and glucose metabolism, potentially mediated through SUMOylation of transcription factors. These findings indicate that Them1 functions not only as an enzymatic regulator but also as a transcriptional regulator of thermogenesis, expanding the current understanding of its role in brown adipocytes and highlighting its potential in therapies for obesity and metabolic disease.

Keywords: non-shivering thermogenesis, brown adipocyte, Them1, LPC, nuclear translocation, bulk RNA-seq, Ingenuity pathway analysis, RT-qPCR, time-course

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1 Introduction

Brown adipose tissue (BAT) is essential for keeping a stable internal body temperature through non-shivering thermogenesis, a process that releases chemical energy as heat. When exposed to cold temperatures, the sympathetic nervous system releases norepinephrine, activating the β_3-adrenergic signaling pathway in brown adipocytes. This signaling cascade activates protein kinase A (PKA), stimulating lipolysis to release free fatty acids that activate uncoupling protein 1 (*Ucp1*) [1]. Ucp1 creates a proton leak that uncouples respiration from ATP synthesis and generates heat. Since thermogenesis uses up a large supply of fatty acids, brown adipocytes heavily regulate this process to prevent depletion of energy reserves [2]. One of the key proteins involved in this regulation is thioesterase superfamily member 1 (Them1), a long-chain fatty acyl-CoA thioesterase that converts the fatty acids needed for thermogenesis into an unusable form, conserving energy (Fig. 1) [1, 3, 4].

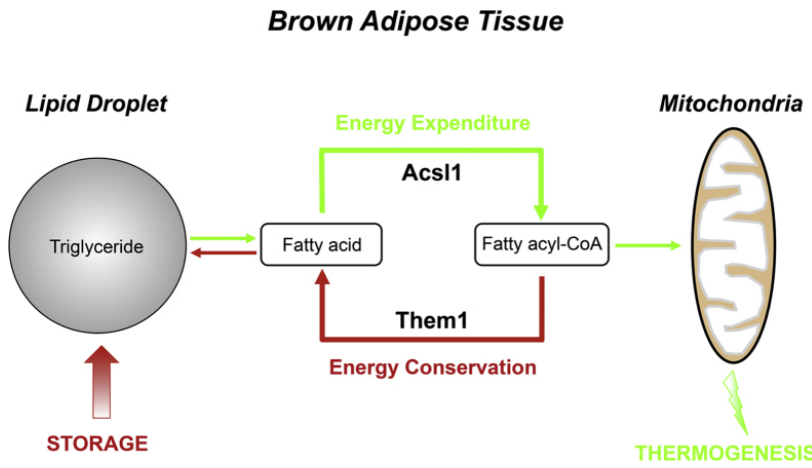


Figure 1: Them1 functions as a regulator for thermogenesis, converting fatty acyl-CoA into fatty acids that are not suitable for energy expenditure. (Okada *et al.*) [4].

Under normal conditions, Them1 forms biomolecular condensates also called puncta near lipid droplets and mitochondria. When in puncta, Them1 is active and hydrolyzes long-chain fatty acyl-CoAs into free fatty acids and coenzyme A, which cannot be catabolized through thermogenesis [4, 5, 6]. However, in response to cold temperatures or nutritional excess, the β_3-

adrenergic signaling pathway causes the downstream activation of protein kinase Cβ (PKCβ), which phosphorylates serine residues at the N-terminus of Them1. Phosphorylation causes Them1 to dissociate from puncta, resulting in Them1 diffusing throughout the cytoplasm (Fig. 2) [1, 7]. Diffuse Them1 is less able to suppress fatty acid oxidation in thermogenesis, allowing the cells to produce more heat.

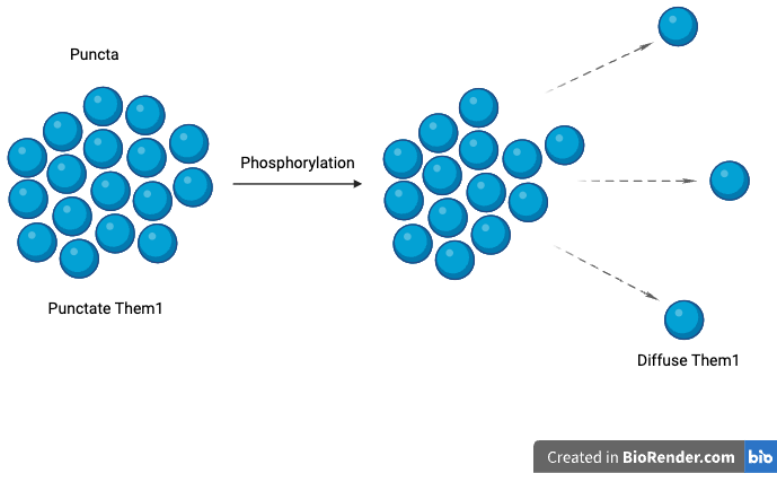


Figure 2: After phosphorylation of punctate Them1, the proteins dissociate from the puncta and diffuse throughout the cytoplasm. The image was created in BioRender.

Once phosphorylation is induced by β_3-adrenergic stimulation, diffuse Them1 is able to translocate into the nucleus [8]. Through a long-term cold-stimulated mouse model, nuclear Them1 was shown to interact with transcription factor carbohydrate response element binding protein (ChREBP), reducing the expression of genes involved in glycolysis and *de novo* lipogenesis [3, 7]. These findings suggest that Them1 functions not only as a regulator of fatty acid oxidation but also as a potential transcriptional regulator in response to β_3-adrenergic stimulation. Building on this in vivo evidence, further in vitro work is needed to elucidate the mechanisms behind Them1's interactions with transcriptional factors as well as to determine the magnitude of nuclear Them1's effect on broader gene expression in brown adipocytes.

To better understand Them1's role as a transcriptional regulator, we investigated how its nuclear localization influences gene expression in brown adipocytes. However, relying on β_3-adrenergic stimulation to induce Them1 phosphorylation would result in numerous metabolic effects that would confound the gene expression results. To bypass this issue, we engineered Them1 variants to mimic the different Them1 localization states. We then compared transcriptomes across three cell groups: those expressing a Them1 mutant that mimics diffuse, phosphorylated Them1 (Them1-DDD); cells expressing a mutant that remains punctate (Them1-AAA); and cells expressing a control construct protein (eGFP) (Fig. 3) [1].

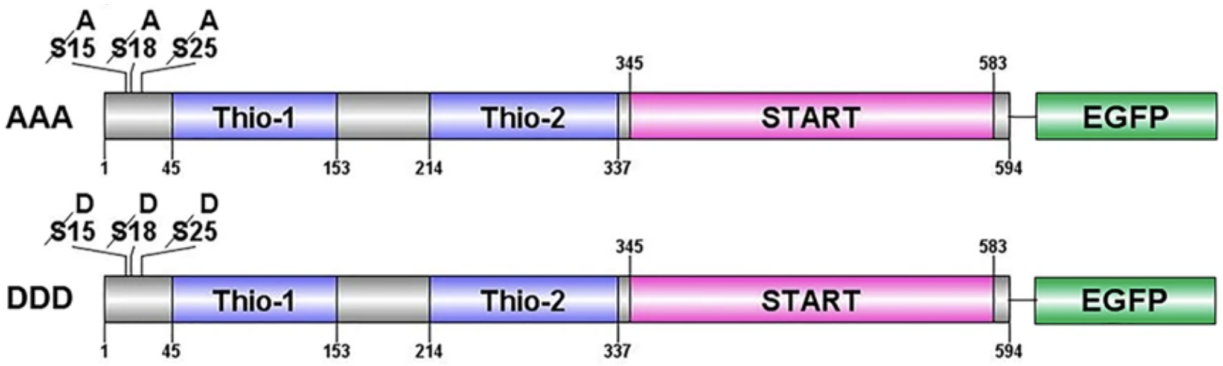


Figure 3: Them1 mutants were engineered as described in [1]. Them1-AAA was engineered by mutating Ser15, Ser18, and Ser25 to Alanine (S15A, S18A, and S25A). Them1-DDD was engineered by mutating Ser15, Ser18, and Ser25 to aspartate (S15D, S18D, and S25D). Both mutants were designed to be fused to eGFP for the purposes of confocal imaging in prior experiments. (Li *et al.*) [1].

In vitro, Them1-DDD alone showed low levels of nuclear trafficking, but when treated with LPC, nuclear trafficking substantially increased [7]. This is explained by Them1's structural biology. Them1 is comprised of tandem enzymatic thioesterase domains and C-terminal lipid-binding (START) domain (Fig. 4, unpublished work). Binding of the lipid lysophosphatidylcholine (LPC) to the C-terminal START domain of Them1 allosterically inhibits Them1 activity and drives a conformational change that exposes a nuclear localization signal (NLS), allowing diffuse Them1 to enter the nucleus (Fig. 4, unpublished work) [9, 7].

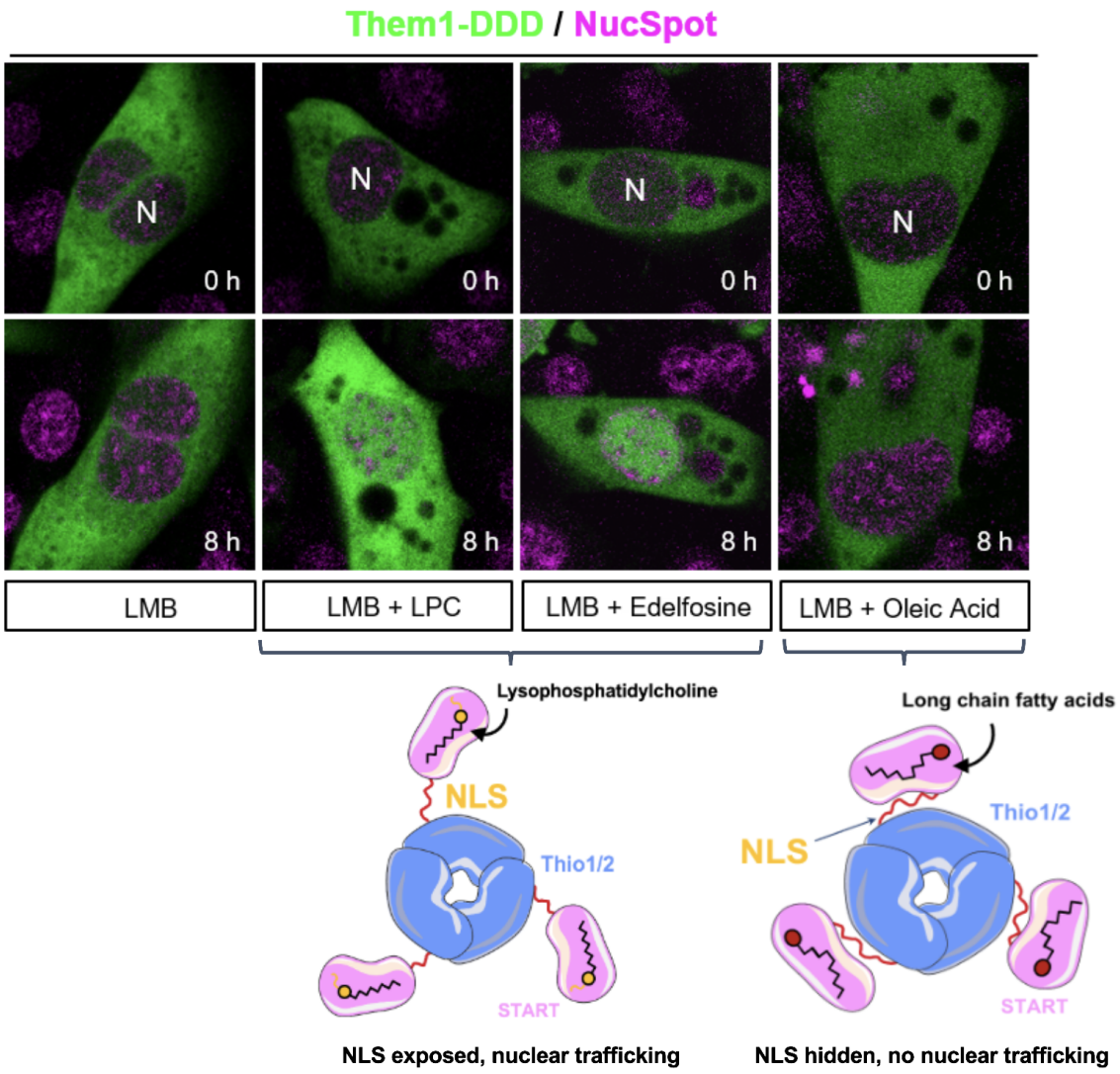


Figure 4: Cells expressing Them1-DDD treated with leptomycin B (LMB), a nuclear transport blocker, alone or in combination with indicated treatments, were visualized with fluorescence imaging. A) LMB alone: Them1-DDD fails to accumulate in the nucleus. B) LMB + LPC: Them1-DDD nuclear accumulation is substantially increased compared to LMB alone. C) LMB + edelfosine (alkyl-lysophospholipid analog structurally similar to LPC): nuclear accumulation is also substantially increased. Schematic below panels B-C illustrates the mechanism behind Them1 nuclear entry: Them1 forms a homotrimer with a thioesterase domain core (Thio 1/2) and flanking START domains attached to the core via a hinge region, and LPC/edelfosine binding reveals the NLS [9, 10]. D) LMB + oleic acid (long-chain fatty acid that stabilizes the Them1 trimer): Them1-DDD fails to accumulate in the nucleus, consistent with below schematic depicting the hidden NLS. This figure contains unpublished data from the Hagen Lab.

This work aims to expand the current understanding of Them1 beyond its established function in regulating fatty acid metabolism and thermogenesis by examining how LPC binding and nuclear translocation affect gene expression. In identifying pathways associated with Them1 nuclear trafficking, we offer new insight into the molecular mechanisms that coordinate thermogenesis and metabolism in brown adipose tissue.

2 Materials and Methods

2.1 Brown Adipocyte Culture

Immortalized brown adipocytes (iBAs) were derived from murine BAT for cell culture [11]. Isolated BAT cells no longer express Them1 due to the absence of physiological signals that stimulate Them1 expression, allowing for controlled expression of each Them1 mutant [1]. Standardized methods for iBA cell culture were used. Cells were cultured in 2× Dulbecco's Modified Eagle Medium (DMEM) without sodium pyruvate, 20% Fetal Bovine Serum (FBS) growth medium, and 1% penicillin-streptomycin solution (P/S). Cells were preserved under standard culture conditions and passaged once they reached 80% confluence to maintain exponential growth. Cells were then cultured until they reached 100% confluence and remained in culture for an additional 24 h before differentiation. At this stage, the culture medium was replaced with an induction medium consisting of DMEM, 10% FBS, 1% P/S, 0.02 μM insulin, 1 nM triiodothyronine (T3), 1 μM rosiglitazone, 2 μg/mL dexamethasone, 125 μM indomethacin, and 500 μM isobutylmethylxanthine (IBMX).

Approximately 48 h after induction, when intracellular lipid droplets were visible, cells were transduced with recombinant adenoviruses expressing Them1-DDD, Them1-AAA, or eGFP control at a multiplicity of infection (MOI) of 40, since this MOI has been shown to produce Them1 expression comparable to that in BAT during cold exposure [1]. The Them1 adenovirus was created by ViraQuest. All adenoviral vector work was done as described in [1] by my lab mentor under appropriate recombinant DNA/RNA guidelines.

2.2 Bulk RNA Sequencing

To observe differences in gene expression between cells with nuclear Them1 and cells with punctate Them1, bulk RNA sequencing (RNA-seq) was performed on cells expressing the Them1-DDD and Them1-AAA mutants. Them1-DDD itself undergoes low-level nuclear trafficking, which likely occurs with basal levels of LPC, so we considered the endogenous LPC to be sufficient to cause Them1 nuclear translocation [7]. RNA sequencing was conducted at the Dana-Farber Cancer Institute (DFCI). RNA samples were isolated using RNeasy Mini Kit (QIAGEN) with n = 3, and RNA-seq was performed at 40 million reads per sample using NovaSeq 6000 (Illumina). Gene alignment to the mouse genome and generation of log_2-transformed expression values from raw read counts were performed at the DFCI. Differential expression analysis was performed using DESeq2.

2.3 Ingenuity Pathway Analysis

RNA-seq data from cells expressing Them1-DDD and Them1-AAA was analyzed to identify pathways associated with Them1 localization in cells. The top 3,000 genes upregulated in cells expressing Them1-DDD relative to cells expressing Them1-AAA ($|log_2FC| \geq 0.02$) were compared against the top 3,000 genes downregulated in Them1-DDD relative to Them1-AAA ($|log_2FC| \geq 0.02$). Pathway enrichment analysis based on differential gene expression was conducted using Ingenuity Pathway Analysis (QIAGEN) [12]. Z-score comparison analysis was conducted through IPA to determine pathways with differences in activity between the two cell groups (z-score cutoff = 2). Pathway analysis focused primarily on metabolic pathways, which were most likely to be influenced by nuclear Them1.

2.4 RT-qPCR Analysis

2.4.1 RT-qPCR without LPC Treatment

Reverse transcription-quantitative polymerase chain reaction (RT-qPCR) was performed to validate that the pathways identified as enriched by IPA were influenced by Them1 nuclear translocation. iBAs were cultured, differentiated, and transduced as described in Methods 2.1. RNA was extracted from the treated cells using the Qiagen RNeasy Mini Kit (Qiagen), and RNA concentration was measured with a NanoDrop 2000 spectrophotometer (Thermo Fisher). For each sample, 1 μg RNA was reverse transcribed into cDNA using the SuperScript III First-Strand Synthesis System (Invitrogen) in a 20 μL reaction. For each Them1 cell group (Them1-DDD, Them1-AAA, and control eGFP), two biological replicates of cDNA were generated, with each replicate run in technical duplicate. Each sample was combined with PowerUp SYBR Green Master Mix (Thermo Fisher) and forward and reverse primers for each gene studied. Duplicate wells were run for each unique combination of cDNA sample and primer to reduce the effect of pipetting error on the results. RT-qPCR was performed with the QuantStudio 5 Real-Time PCR System (Thermo Fisher).

Primers were selected to quantify the expression of genes central to thermogenesis, glycolysis, fatty acid oxidation, and lipid biosynthesis. Genes were also selected to be specialized to their respective pathways, avoiding genes with broader roles whose expression changes could be misattributed solely to the identified pathway. The genes selected for primers and their function are shown in Table 1.

Gene Name	**Pathway Name**	**Gene Function**
Ucp1	Thermogenesis	Encodes the mitochondrial uncoupling protein responsible for heat production.
Pgc1a	Thermogenesis	Encodes a major transcriptional coactivator that controls adaptive thermogenesis.
Elovl3	Thermogenesis	Encodes an endoplasmic reticulum enzyme that extends long-chain fatty acids into very long-chain fatty acids.
Srebp1c	Lipogenesis	Encodes a major transcription factor that controls lipid synthesis in response to dietary signals from insulin.
Scap	Lipogenesis	Encodes a major transcription co-factor for lipogenesis.
Acly	Lipogenesis	Encodes the protein that transforms citrate to the acetyl-CoA needed for fatty acid synthesis.
Fasn	Lipogenesis	Encodes the primary enzyme responsible for *de novo* lipogenesis.
Acc1	Lipogenesis	Encodes an enzyme involved in the rate-limiting step in *de novo* fatty acid synthesis.
Adolc	Glucose metabolism	Encodes the enzyme that helps break down sugar inside cells.
Pfkm	Glucose metabolism	Encodes part of an enzyme that plays a critical role in regulating the rate of glycolysis.
G6pdx	Glucose metabolism	Encodes an enzyme that regulates glycolysis.

Table 1: Genes shown in the table were analyzed in RT-qPCR experiments. All gene information came from the NCBI gene search tool [2].

Gene selection was further informed by RNA-seq data, in which chosen genes showed differential expression across cells expressing the Them1-DDD and Them1-AAA mutants relative to eGFP control. *Actb* was amplified with these genes in a separate reaction as an internal control, since *Actb* is a housekeeping gene that encodes a protein crucial to cytoskeleton formation, cell movement, and intracellular signaling [2]. qPCR C_T values were normalized to *Actb* using the $2^{-\Delta\Delta C_T}$ method to determine the relative expression of each gene of interest.

2.4.2 RT-qPCR with LPC Treatment

A limitation with the current experiments is that some differences in gene expression could be caused by differences in metabolic activity between the Them1-AAA and Them1-DDD cell groups. This is demonstrated by the high expression of lipid biosynthesis genes in the Them1-AAA cell group, since Them1-AAA suppresses thermogenesis and lower rates of thermogenesis and fatty acid oxidation would cause cells to direct energy toward forming lipids. To address this limitation, we treated cells with LPC to promote Them1 translocation into the nucleus which may amplify its effects on gene expression. In this way, we could identify the pathways with increased differential activity in the Them1-DDD cell group as being directly influenced by nuclear Them1.

Since LPC is an unstable molecule that degrades quickly, the time that cells are treated with LPC before being analyzed is very important [13]. Measuring gene expression too soon after LPC treatment would not allow transcriptional changes to accumulate, while measuring too late may result in missing the changes, since LPC would already be metabolized and the expression of responsive genes may have normalized. RT-qPCR time-course experiments were conducted to address this concern and observe how different treatment times affect gene expression. Cells were treated with 1, 2, and 4 h of L-α-Lysophosphatidylcholine (25 μM; Sigma-Aldrich). The dosage of LPC treatment was chosen to not compromise cell viability measured by a colorimetric assay using crystal violet. RT-qPCR was conducted as described

in Methods 2.4.1. From this, the time frame that produced the clearest differences between the gene expression of Them1-DDD and control eGFP was determined.

3 Results and Discussion

3.1 RNA-seq Pathway Analysis

Due to high variability in each group, the differential expression of individual genes between groups did not reach statistical significance ($p \geq 0.05$). However, pathway analysis is more robust and revealed several enriched pathways with differences in activity between the two cell groups (z-score cutoff = 2, p-value ≤ 0.05). Pathway analysis first focused on metabolic pathways, since they are the most related to thermogenesis and would likely be impacted by Them1 activity (Fig. 5). Notably, pathways related to fatty acid oxidation were uniquely activated in the Them1-DDD cell group. Pathways related to glycolysis were also uniquely inhibited in the Them1-DDD cell group, consistent with a previous study's findings that the interaction of nuclear Them1 and ChREBP reduces the expression of enzymes mediating glycolysis in brown adipocytes [3].

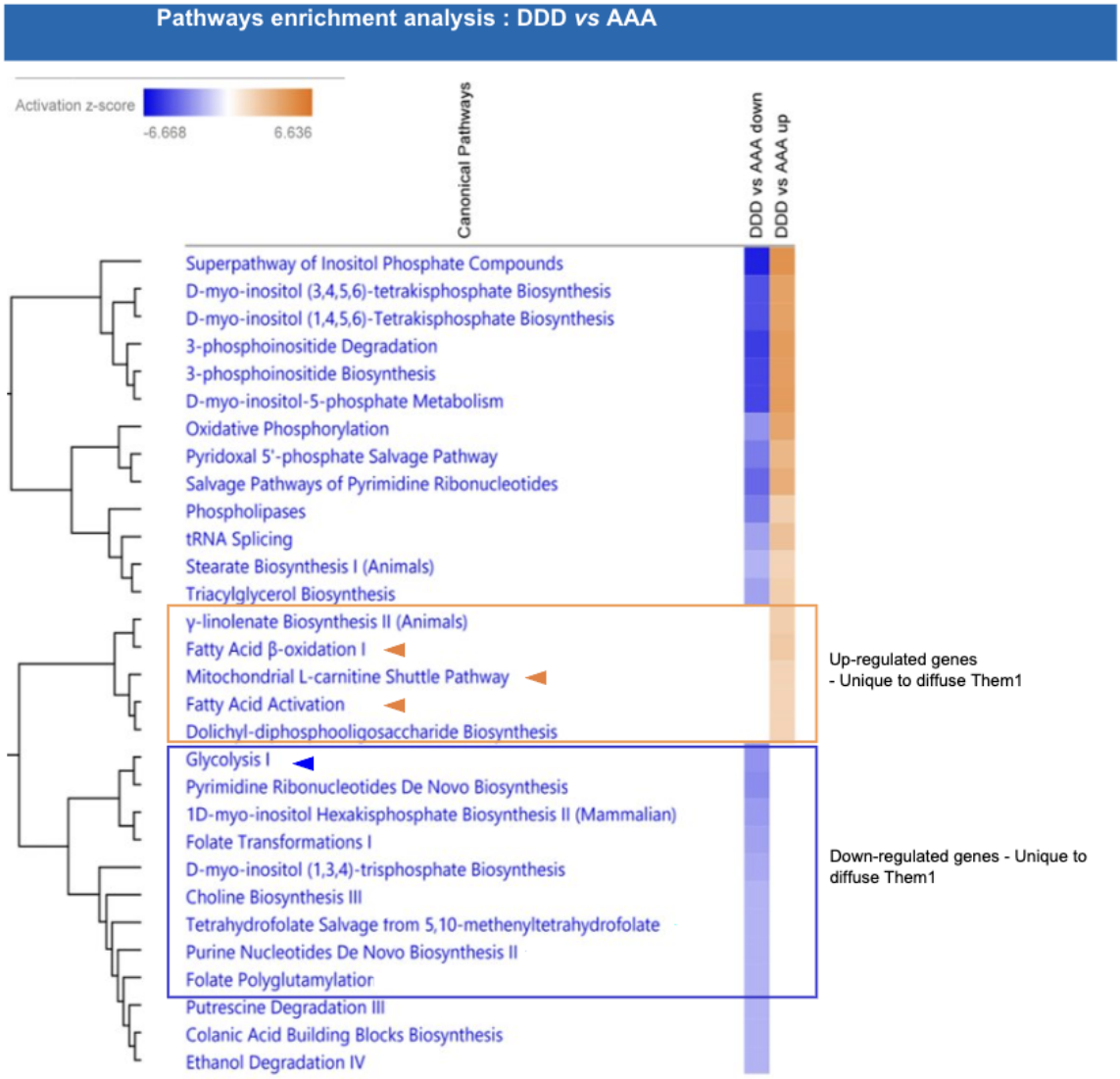


Figure 5: This figure depicts the metabolic pathways enriched after conducting comparison analysis of the top 3,000 genes upregulated in Them1-DDD compared to Them1-AAA cell group and the top 3,000 genes downregulated in Them1-DDD compared to Them1-AAA cell group. Sample size = 3 per group. log_2FC cutoff = 0.02. Differential expression analysis is performed using DESeq2.

In addition, we looked at reactome pathways (Fig. 6), which revealed enriched pathways that validated the metabolic pathway analysis. Pathways related to mitochondrial structure and function, which facilitate fatty acid oxidation, were also uniquely activated in the Them1-

DDD cell group. Pathways related to controlling lipid and cholesterol biosynthesis by the sterol regulatory element-binding protein (SREBP) were uniquely inhibited in the Them1-DDD cell group. While z-score analysis of glucose metabolism pathways indicated that they were both activated and inhibited, further investigation into the Benjamini-Hochberg p-value revealed that the pathways were significantly more inhibited than activated, consistent with the findings on glycolysis from metabolic pathway analysis.

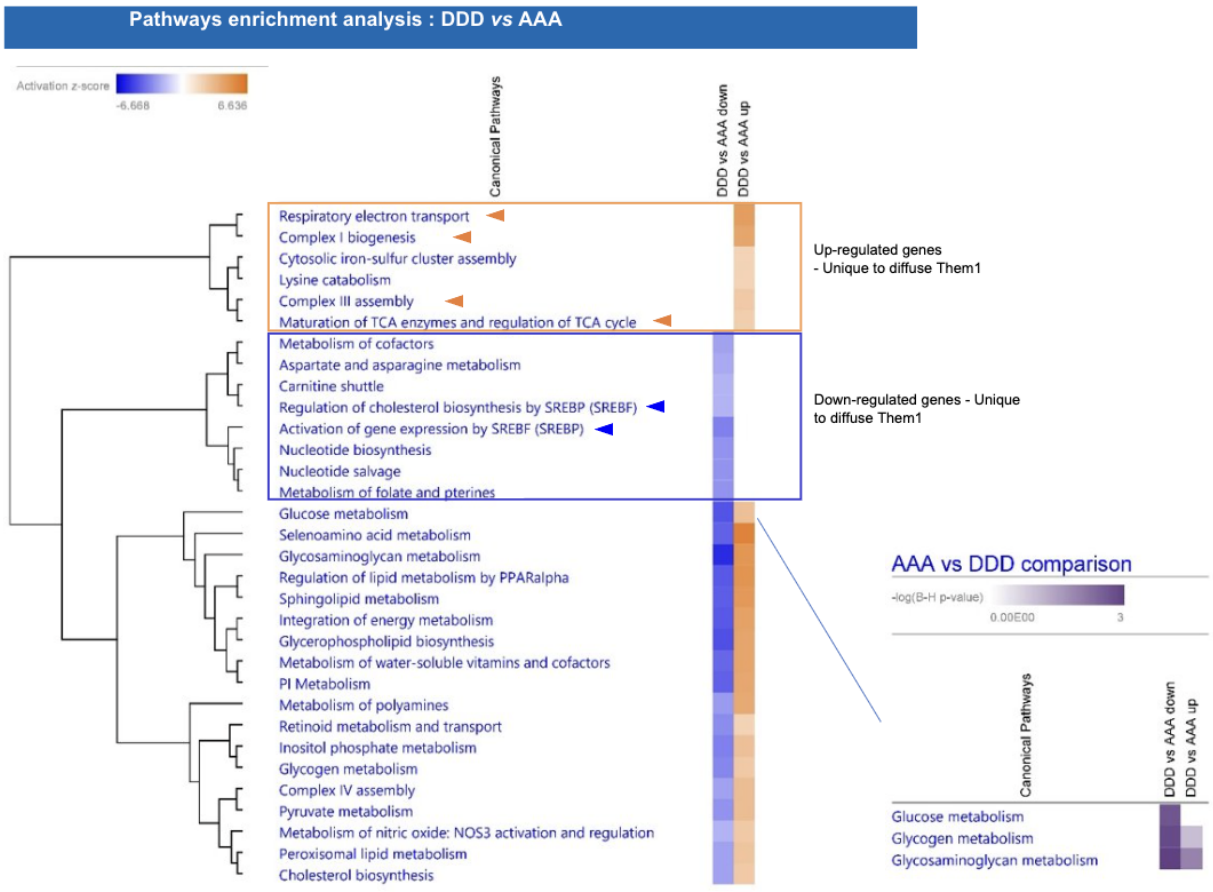


Figure 6: This figure depicts the reactome pathways enriched after conducting comparison analysis of the top 3,000 genes upregulated in Them1-DDD compared to Them1-AAA cell group and the top 3,000 genes downregulated in Them1-DDD compared to Them1-AAA cell group. Sample size = 3 per group. log_2FC cutoff = 0.02. Differential expression analysis is performed using DESeq2.

Beyond pathways with established roles in thermogenesis, our analysis also identified

enrichment of processes with less well-characterized connections to thermogenesis. The SUMOylation pathway (Fig. 7A), which causes the covalent attachment of small ubiquitin-like modifier (SUMO) proteins to target substrates, was found to be more active in cells expressing Them1-DDD. This process changes protein stability and function, and is crucial to the regulation of gene expression and cellular stress responses [14, 15].

We also observed enrichment of circadian clock pathways, which were more active in cells expressing Them1-DDD than in cells expressing Them1-AAA (Fig. 7B). During acute cold exposure, organisms prioritize heat production, which can disrupt normal circadian rhythms as energy is focused toward sustaining thermogenesis [16, 17]. Since Them1 accumulates in the nucleus following cold exposure, we hypothesize that, once the organism has adapted to prolonged cold exposure, nuclear Them1 helps curtail thermogenesis while helping to restore the normal circadian clock.

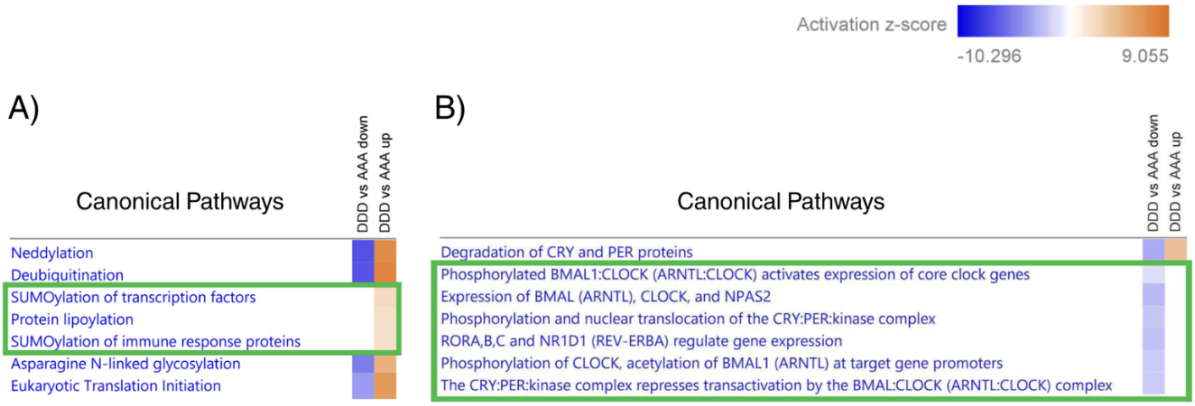


Figure 7: Each panel depicts the canonical pathways enriched after conducting comparison analysis of the top 3,000 genes upregulated in Them1-DDD compared to Them1-AAA cell group and the top 3,000 genes downregulated in Them1-DDD compared to Them1-AAA cell group. A) depicts pathways associated with metabolism of proteins and B) depicts pathways linked to the circadian rhythm. Sample size = 3 per group. log_2FC cutoff = 0.02. Differential expression analysis is performed using DESeq2.

3.2 RT-qPCR Pathway Validation

To validate the results from RNA-seq analysis, RT-qPCR was conducted on cells expressing Them1-DDD, Them1-AAA, and control Ad-eGFP, focusing on genes related to metabolic pathways. Based on the expression data collected, we found a trend that nearly all genes were downregulated in the Them1-DDD cell group compared with the control and were upregulated in the Them1-AAA cell group when compared with the control (Fig. 8). We hypothesize that this is a result of Them1's activity in the nucleus.

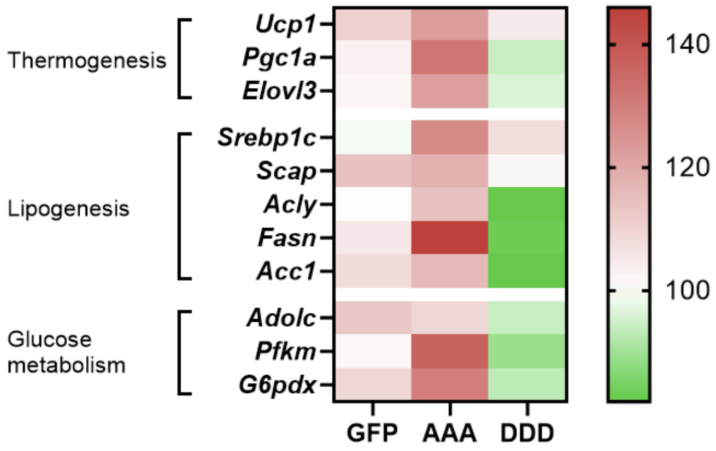


Figure 8: This figure depicts the expression level for different genes in the Them1-AAA, Them1-DDD, and eGFP control groups as a heatmap. n = 2 per group.

This hypothesis aligns with the findings from our pathway analysis on SUMOylation activity. Since SUMOylation of transcription factors is known to be linked to transcriptional repression, its enrichment in the Them1-DDD cell group is consistent with the broader pattern of transcriptional repression observed in this group relative to Them1-AAA [15]. This suggests that SUMOylation of transcription factors may be the mechanism by which nuclear Them1 contributes to the repressive gene regulation associated with the Them1-DDD cell group.

3.3 RT-qPCR of LPC-treated Cells

The goal of conducting RT-qPCR time-course experiments on LPC-treated cells was to determine whether LPC treatment expanded gene expression differences by increasing Them1 nuclear trafficking. The time-course experiments indicated that 2 h is the optimal LPC treatment time frame to observe the effects of LPC-induced Them1 nuclear translocation. This is depicted in Fig. 9, where the differences in Ucp1 and Scap expression between the eGFP control and Them1-DDD groups became more pronounced over time, peaking or plateauing around 2 h before diminishing by 4 h of treatment. A similar trend is seen among other genes that were tested. However, some genes such as *Acly* and *Acc1* showed the largest expression differences with no LPC treatment. We suspect that this may be due to LPC acting as a signal molecule in the cells, meaning that no LPC treatment is not indicative of increased effects of nuclear Them1 on gene expression. It has been shown that LPC formation induces thermogenesis in brown adipocytes, which would explain the observed decrease in *Acly* and *Acc1* expression, since these genes are involved in lipid biosynthesis, which is downregulated in response to increased thermogenesis. This would cause the observed difference in expression at 2 h LPC treatment to be smaller than that with no treatment. Time-course data for all genes tested are provided in Appendix A.

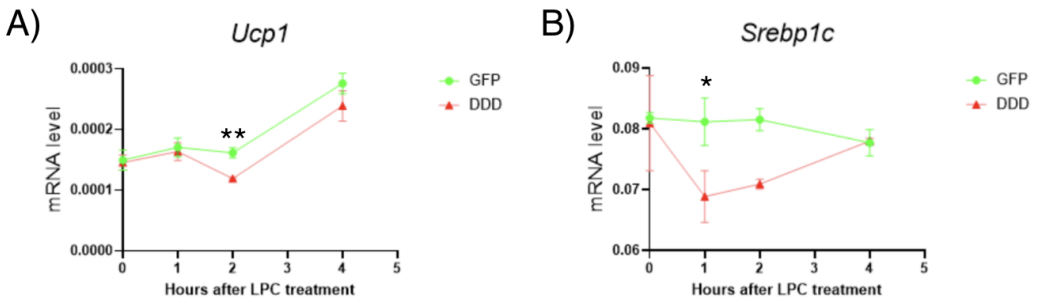


Figure 9: Results from LPC time-course experiments. Each panel depicts the mRNA expression of a different gene over 4 h of LPC treatment times for two cell groups expressing Them1-DDD and eGFP control. n = 2 per group for each time point. Error bars represent standard deviation (SD). A two-way ANOVA and multiple comparisons were performed on the time-course data. * indicates p-value < 0.05 when comparing gene expression between eGFP vs Them1-DDD. ** indicates p-value < 0.01 when comparing gene expression between eGFP vs Them1-DDD.

Based on the time-course results of the Them1-AAA cell group, we noticed that LPC treatment had a dramatic effect on the expression of many genes, especially for genes related to lipid biosynthesis (*Srebp1c*, *Scap*, *Fasn*, *Acly*, *Acc1*). This is highlighted in the graph of *Srebp1c* (Fig. 10B), where *Srebp1c* expression substantially decreased after 1 h LPC treatment. This drastic change was unexpected since LPC treatment should not induce Them1-AAA nuclear translocation and thus should not result in altered gene expression. This result may also be due to LPC inducing thermogenesis in the cells and causing the expression of genes related to lipid biosynthesis to decrease [18, 19].

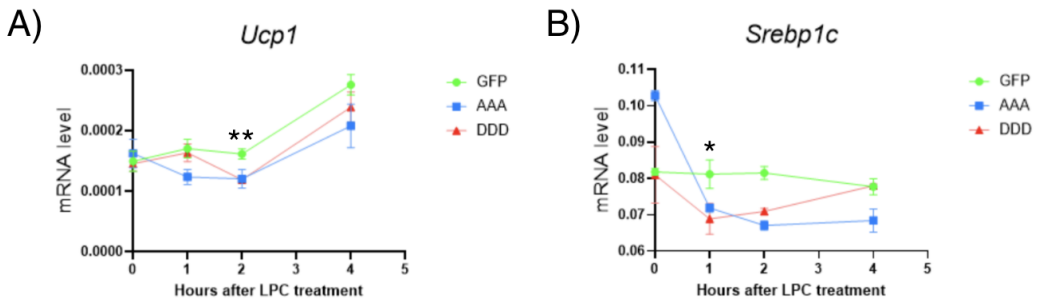


Figure 10: Results from LPC time-course experiments. Each panel depicts the mRNA levels of a different gene over 4 h of LPC treatment times for three cell groups expressing Them1-DDD, Them1-AAA, and eGFP control. n = 2 per group for each time point. Error bars represent standard deviation (SD). A two-way ANOVA and multiple comparisons were performed on the time-course data. * indicates p-value < 0.05 when comparing gene expression between eGFP vs Them1-DDD. ** indicates p-value < 0.01 when comparing gene expression between eGFP vs Them1-DDD.

A two-way ANOVA and multiple comparisons were performed on the time-course data for each gene. *Ucp1* showed a statistically significant difference in expression between Them1-DDD cells and eGFP control at 2 h (p-value < 0.01). *Srebp1c* showed a statistically significant difference at 1 h (p-value < 0.05). All other time points for both genes showed no significant differences in expression between cell groups. However, many other genes only followed a visual trend where gene expression differences between the cell groups peaked or plateaued at 2 h LPC treatment but did not reach statistical significance. This lack of significance may be caused by the low sample size (n = 2) in the time-course experiments.

4 Future Work

There are several directions for future work to strengthen and expand our findings. Our lab plans to collect RNA-seq data from cells expressing Them1-DDD with LPC treatment for 2 h, Them1-DDD with no treatment, and eGFP control to observe the effects of increased nuclear Them1 on gene expression on a global scale. In addition, we plan to collect RNA-seq data from cells expressing Them1-DDD that are treated with edelfosine, an alkyl-lysophospholipid

analog that acts as a non-metabolizable version of LPC [10]. Treatment with edelfosine would allow Them1-DDD nuclear trafficking without activating LPC-mediated signaling pathways, allowing us to determine what transcriptional changes are directly caused by nuclear Them1.

In addition, pathway enrichment should be confirmed through protein analysis. Tests such as Western blotting could confirm whether key molecules in a pathway show changes consistent with the RNA-seq data, since mRNA level differences do not always predict protein-level outcomes. Key pathways we are interested in further exploring are those related to the circadian clock and SUMOylation, since these pathways have no established association with Them1 in the literature, but their biological connections to thermogenesis suggest potential links between the processes. In particular, we plan to investigate the SUMOylation of transcription factors to determine if it could be a potential mechanism behind the major transcriptional repression associated with Them1-DDD. Since the differential expression of the SUMO2 and SUMO3 genes was associated with SUMOylation activity in the Them1-DDD cell group based on IPA analysis, we plan to use the SUMO-2/3 antibody to perform Western blots and immunostaining for confocal microscopy.

These future directions will provide great insight into not only the impact of Them1 nuclear trafficking on gene expression but also the processes behind these transcriptional changes.

5 Conclusion

These findings suggest that nuclear Them1 acts as a transcriptional regulator for thermogenesis, while punctate Them1 simultaneously suppresses thermogenesis, creating a balance that maintains stable heat production.

RNA-seq and pathway analysis identified unique activity in Them1-DDD compared to Them1-AAA in pathways related to fatty acid oxidation, mitochondrial function, glycolysis, and cholesterol and lipid biosynthesis. RT-qPCR corroborated these findings based on the

expression of genes involved in those metabolic pathways. These two approaches support the idea that Them1's role goes beyond its established function as an enzymatic regulator to include a role as a transcriptional regulator.

RT-qPCR experiments conducted both with and without LPC treatment demonstrated a consistent trend of downregulation in genes related to thermogenesis, lipogenesis, and glucose metabolism in the Them1-DDD cell group. We posit that this transcriptional repression in Them1-DDD cells is caused by the SUMOylation of transcription factors, which was uniquely activated in Them1-DDD according to RNA-seq IPA. This suggests that during nuclear trafficking, Them1 may activate SUMOylation pathways as a way to perform its function as a transcriptional regulator.

In addition to pointing out potential effects of nuclear Them1 on gene expression and the mechanisms behind them, these findings have applications in therapeutics. Them1 knockout in mouse models has been shown to cause resistance to obesity by increasing lipid catabolism, which mitigates excess fat storage [1, 20]. Validating the transcriptional function of Them1 would help develop new treatments targeting Them1 for diseases caused by excess of fat storage, such as obesity, with Them1 acting as both a transcriptional and enzymatic regulator for thermogenesis.

6 Key Takeaways

This research suggests that Them1 acts not only as an enzyme suppressing thermogenesis, but also as a transcriptional regulator when trafficking in and out of the nucleus. Given that loss of Them1-mediated thermogenesis suppression has been shown to protect against obesity in mouse models, confirming this additional function of Them1 expands its potential in treatments of obesity and other metabolic diseases. This study also identified pathways with no existing connections to Them1 as potentially influenced by nuclear Them1 activity, opening up more avenues of research into how thermogenesis interacts with other cellular

processes.

7 Acknowledgments

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A Time-Course Results

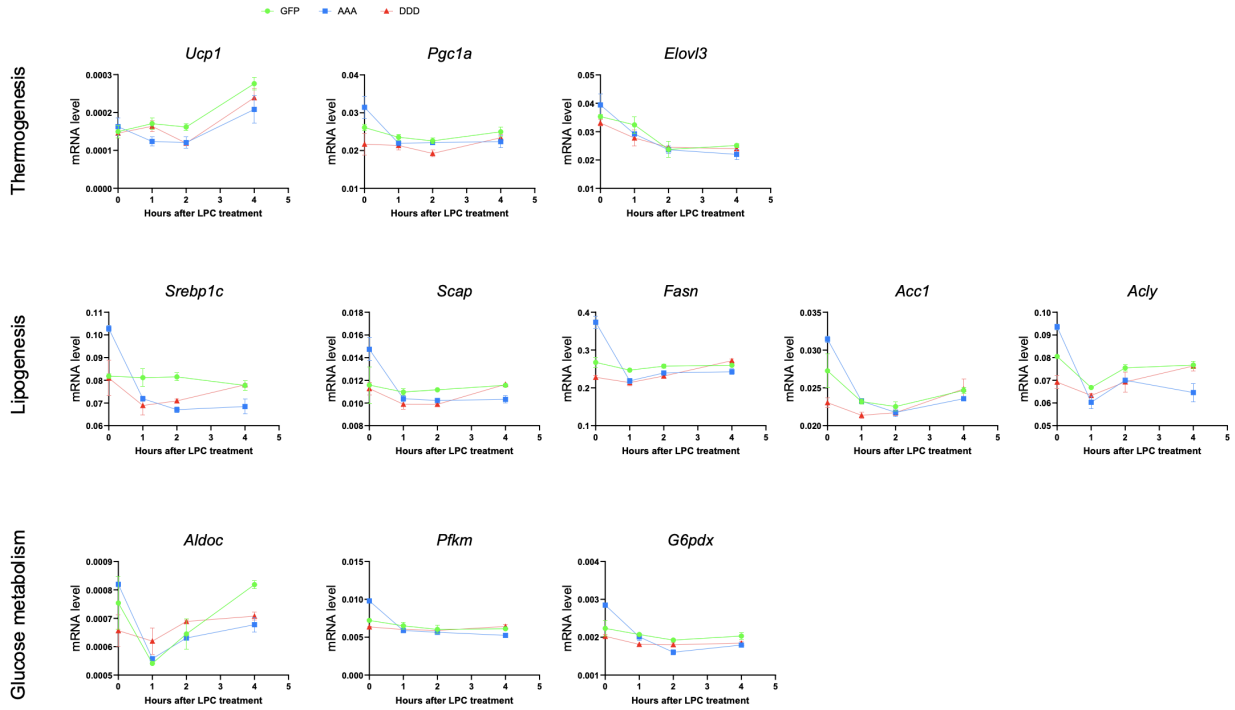


Figure 11: Results from LPC time-course experiments. Each graph depicts the mRNA levels of different genes over different LPC treatment times for three cell groups expressing Them1-AAA, Them1-DDD, and Them1-GFP. The genes are grouped by the pathway they are involved in. n = 2 per group for each time point. Error bars represent standard deviation (SD).